

Programme: B E – Information Science & Engineering
Internal Assessment – I

TERM : 15 April 2024-27 July 2024		COURSE NAME: Design and Analysis of Algorithms
DATE : 31-05-2024	TIME:9:30am-10:30am	COURSE CODE : IS43
MAX MARKS: 30	Semester: 4	PORTIONS : L 1 – L16



Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Consider the following algorithm. Algorithm Mystery (A [0.....n-1]) //Input: An array A[0.....n-1] for i←0 to n-2 do for j←i+1 to n-1 do if A[i]=A[j] return false return true i. What does this algorithm compute? ii. What is its basic operation? iii. How many times is the basic operation is executed? iv. What is the efficiency class of this algorithm?	L3	CO1	7
b	Define divide and conquer technique. Apply Quick sort for the following instance of an array [55,22,44,77,33,99,11,88,66]. Compute the best-case and worst-case time complexity.	L3	CO2	8
2.a	Write the general plan for analysing time efficiency of recursive algorithms. Mathematically analyse tower of hanoi and find its time complexity. Construct recursive tree for tower of hanoi for number of discs is 3.	L3	CO1	7
b	Describe brute-force approach. Write a selection sort algorithm. Sort the list T, R, I, A, N, G, L, E in alphabetical order by selection sort.	L3	CO2	8
3.a	Explain the various asymptotic notations with neat graph and examples. Compare order of growth of merge sort and bubble sort.	L4	CO1	7
b	Define space and time trade-off algorithm. Apply Horspool's algorithm to search for the pattern AT_THAT in the text WHICH_FINALLY_HALTS_AT_THAT.	L3	CO2	8

*L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create